

Tasks

1.

Boot up a RHEL 8 virtual machine and perform a custom installation using the graphical installer, making sure to select 'LVM' as the partitioning scheme for the root (/) partition.

>

Procedure

RHEL 8 was booted using the graphical installer. In the Installation Destination section, Custom partitioning was selected. The root / partition was configured using LVM.

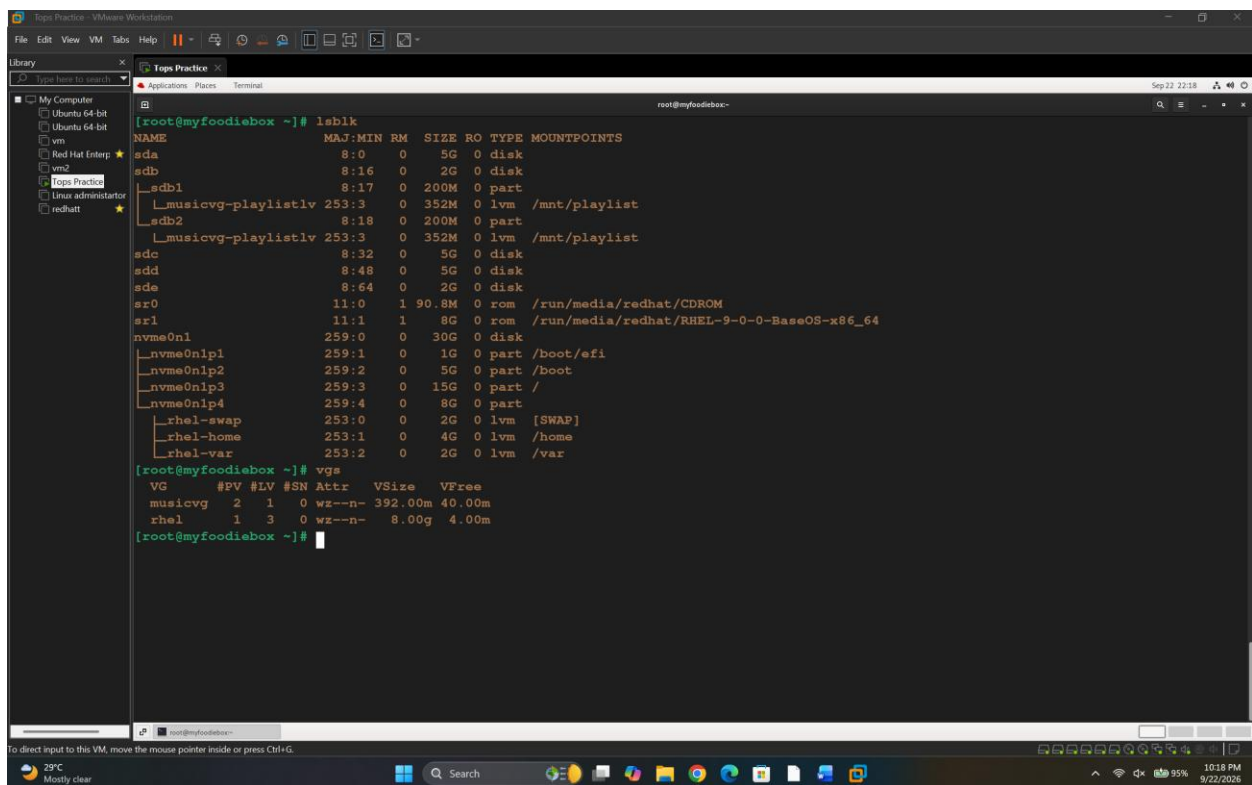
Result

RHEL 8 was successfully installed with the root filesystem managed by LVM.

2.

Inside your RHEL 8 system, use the 'lsblk' and 'vgs' commands to verify that your root partition is managed by LVM and note down the Volume Group and Logical Volume names.

>



```
[root@myfoodiebox ~]# lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                                  8:0    0   5G  0 disk 
sdb                                  8:16   0    2G  0 disk 
sdb1                                 8:17   0   200M 0 part 
└─_musicvg-playlistlv 253:3    0   352M 0 lvm  /mnt/playlist
sdb2                                 8:18   0   200M 0 part 
└─_musicvg-playlistlv 253:3    0   352M 0 lvm  /mnt/playlist
sdc                                  8:32   0    5G  0 disk 
sdd                                  8:48   0    5G  0 disk 
sde                                  8:64   0    2G  0 disk 
sr0                                 11:0    1  90.8M  0 rom   /run/media/redhat/CDCROM
erl                                 11:1    1    8G  0 rom   /run/media/redhat/RHEL-9-0-0-BaseOS-x86_64
nvme0n1                             259:0   0   30G  0 disk 
└─_nvme0n1p1                    259:1   0    1G  0 part /boot/efi
   _nvme0n1p2                    259:2   0    5G  0 part /boot
   _nvme0n1p3                    259:3   0   15G  0 part /
   _nvme0n1p4                    259:4   0    8G  0 part 
└─rhel-swap                     253:0   0    2G  0 lvm  [SWAP]
   rhel-home                     253:1   0    4G  0 lvm  /home
   rhel-var                      253:2   0    2G  0 lvm  /var

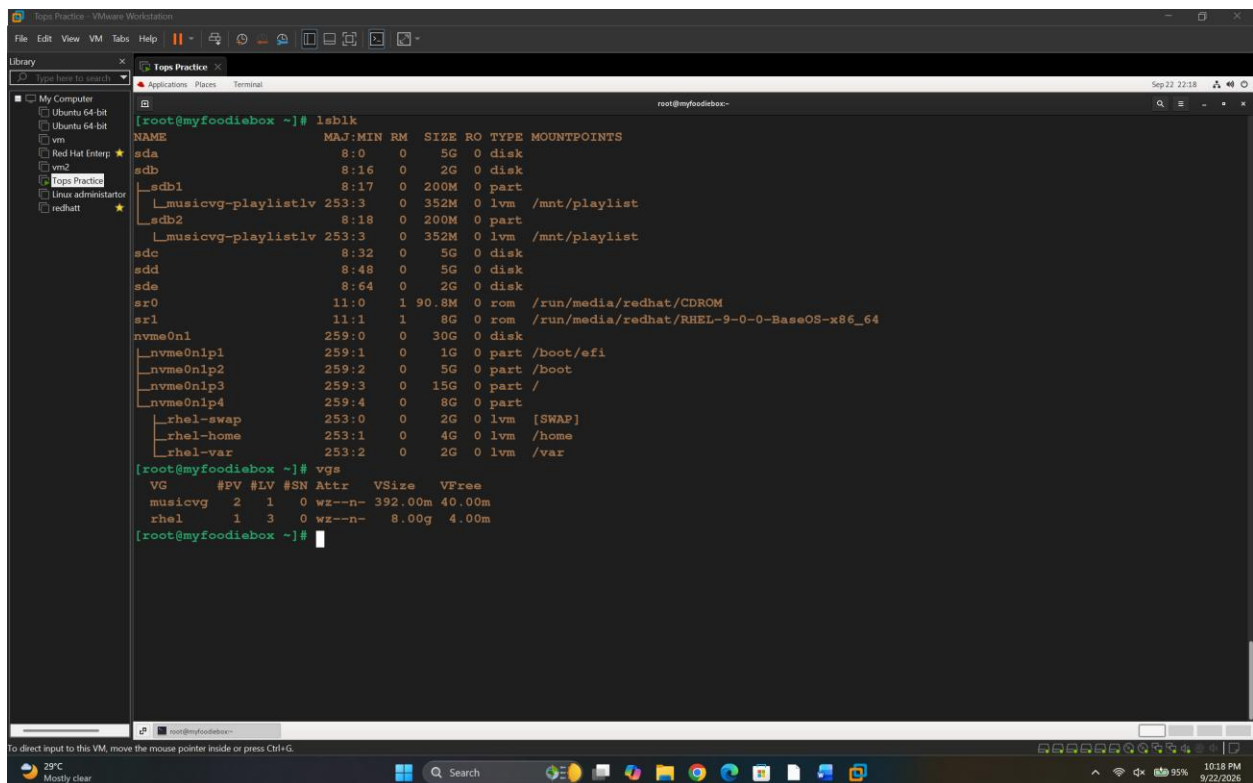
[root@myfoodiebox ~]# vgs
VG      #PV #LV #SN Attr   VSize VFree
musicvg  2   1   0 wz--n- 392.00m 40.00m
rhel     1   3   0 wz--n-  8.00g  4.00m

[root@myfoodiebox ~]#
```

3.

Add a new virtual disk to your RHEL 8 VM, then use 'pvcreate', 'vgextend', and 'lvextend' to increase the size of your existing root Logical Volume by at least 1GB, and resize the filesystem to use the new space. Hint: Use 'df -h' before and after to confirm the size change.

>



```
[root@myfoodiebox ~]# lsblk
NAME                                MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                                  8:0    0   5G  0 disk
sdb                                  8:16    0   2G  0 disk
sdb1                                 8:17    0 200M  0 part
└─musicvg-playlistlv 253:3    0 352M  0 lvm  /mnt/playlist
sdb2                                 8:18    0 200M  0 part
└─musicvg-playlistlv 253:3    0 352M  0 lvm  /mnt/playlist
sdc                                  8:32    0   5G  0 disk
sdd                                  8:48    0   5G  0 disk
sde                                  8:64    0   2G  0 disk
sr0                                  11:0    1 90.8M  0 rom  /run/media/redhat/CDFROM
sr1                                  11:1    1   8G  0 rom  /run/media/redhat/RHEL-9-0-0-BaseOS-x86_64
nvme0n1                             259:0    0 30G  0 disk
├─nvme0n1p1                         259:1    0   1G  0 part /boot/efi
├─nvme0n1p2                         259:2    0   5G  0 part /boot
├─nvme0n1p3                         259:3    0  15G  0 part /
├─nvme0n1p4                         259:4    0   8G  0 part
├─rhel-swap                         253:0    0   2G  0 lvm  [SWAP]
├─rhel-home                         253:1    0   4G  0 lvm  /home
└─rhel-var                         253:2    0   2G  0 lvm  /var

[root@myfoodiebox ~]# vgs
VG      #PV #LV #SN Attr   VSize   VFree
musicvg  2   1   0 wz--n- 392.00m 40.00m
rhel     1   3   0 wz--n-  8.00g  4.00m

[root@myfoodiebox ~]#
```

4.

Create a new Logical Volume called 'musicdata' of 500MB in your existing Volume Group, format it with ext4, mount it at /mnt/music, and add an entry to /etc/fstab so it mounts automatically on boot. Constraint: Use only command-line tools (no GUI) for this task.

>

```

Tops Practice - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
  Ubuntu 64-bit
  Ubuntu 64-bit
  vm
  Red Hat Enterp
  vm2
  Tops Practice
  Linux administrator
  redhatt
Tops Practice
Applications Places Terminal
root@myfoodiebox:~# lsblk
sr0              11:0    1 90.8M 0 rom  /run/media/redhat/CDROM
sr1              11:1    1  8G 0 rom  /run/media/redhat/RHEL-9-0-0-BaseOS-x86_64
nvme0n1          259:0    0 30G 0 disk
├─nvme0n1p1      259:1    0  1G 0 part  /boot/efi
├─nvme0n1p2      259:2    0   5G 0 part  /boot
├─nvme0n1p3      259:3    0  15G 0 part  /
├─nvme0n1p4      259:4    0   8G 0 part
├─rhel-swap      253:0    0   2G 0 lvm  [SWAP]
├─rhel-home      253:1    0   4G 0 lvm  /home
└─rhel-var       253:2    0   2G 0 lvm  /var

[root@myfoodiebox ~]# vgs
VG                #PV #LV #SN Attr   VSize  VFree
musicvg          2   1   0 wz--n- 392.00m 40.00m
rhel              1   3   0 wz--n-   8.00g  4.00m

[root@myfoodiebox ~]# pvcreate /dev/sdb
Cannot use /dev/sdb: device is partitioned

[root@myfoodiebox ~]# pvcreate /dev/sdc
Physical volume "/dev/sdc" successfully created.

[root@myfoodiebox ~]# vgextend rhel /dev/sdb
Cannot use /dev/sdb: device is partitioned
Command requires all devices to be found.

[root@myfoodiebox ~]# vgextend rhel /dev/sdc
Volume group "rhel" successfully extended

[root@myfoodiebox ~]# vgs
VG                #PV #LV #SN Attr   VSize  VFree
musicvg          2   1   0 wz--n- 392.00m 40.00m
rhel              2   3   0 wz--n-  13.00g   5.00g

[root@myfoodiebox ~]# df -h
Filesystem          Size  Used Avail Use% Mounted on
devtmpfs             844M   0 844M   0% /dev
tmpfs                875M   0 875M   0% /dev/shm
tmpfs               350M  9.8M 340M   3% /run
/dev/nvme0n1p3       15G   3.8G 12G   25% /
/dev/nvme0n1p2       5.0G 282M  4.8G   6% /boot
/dev/nvme0n1p1     1022M   7.0M 1015M   1% /boot/efi
/dev/mapper/rhel-home 4.0G   66M   4.0G   2% /home
/dev/mapper/rhel-var  2.0G  203M   1.8G  10% /var

To direct input to this VM, move the mouse pointer inside or press Ctrl+G
29°C
Mostly clear
Sep 22 22:32
10:33 PM
9/22/2026
```

```

Tops Practice - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
  Ubuntu 64-bit
  Ubuntu 64-bit
  vm
  Red Hat Enterp
  vm2
  Tops Practice
  Linux administrator
  redhatt
Tops Practice
Applications Places Terminal
root@myfoodiebox:~# vgextend rhel /dev/sdc
Volume group "rhel" successfully extended

[root@myfoodiebox ~]# vgs
VG                #PV #LV #SN Attr   VSize  VFree
musicvg          2   1   0 wz--n- 392.00m 40.00m
rhel              2   3   0 wz--n-  13.00g   5.00g

[root@myfoodiebox ~]# df -h
Filesystem          Size  Used Avail Use% Mounted on
devtmpfs             844M   0 844M   0% /dev
tmpfs                875M   0 875M   0% /dev/shm
tmpfs               350M  9.8M 340M   3% /run
/dev/nvme0n1p3       15G   3.8G 12G   25% /
/dev/nvme0n1p2       5.0G 282M  4.8G   6% /boot
/dev/nvme0n1p1     1022M   7.0M 1015M   1% /boot/efi
/dev/mapper/rhel-home 4.0G   66M   4.0G   2% /home
/dev/mapper/rhel-var  2.0G  203M   1.8G  10% /var
tmpfs               175M  100K 175M   1% /run/user/1000
/dev/sr1             8.0G   8.0G   0 100% /run/media/redhat/RHEL-9-0-0-BaseOS-x86_64
/dev/sr0             91M    91M   0 100% /run/media/redhat/CDROM
/dev/mapper/musicvg-playlistlv 323M  2.1M  300M   1% /mnt/playlist

[root@myfoodiebox ~]# lvextend -L +1G /dev/rhel/root
Logical volume root not found in volume group rhel.
[root@myfoodiebox ~]# lvextend -L +1G /dev/mapper/rhel-root
Logical volume root not found in volume group rhel.

[root@myfoodiebox ~]# lvs
LV                VG                Attr   LSize  Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
playlistlv        musicvg          -wi-ao---- 352.00m
home              rhel              -wi-ao----  4.00g
swap              rhel              -wi-ao----  2.00g
var               rhel              -wi-ao----  2.00g

[root@myfoodiebox ~]# vgs
VG                #PV #LV #SN Attr   VSize  VFree
musicvg          2   1   0 wz--n- 392.00m 40.00m
rhel              2   3   0 wz--n-  13.00g   5.00g

[root@myfoodiebox ~]#
[root@myfoodiebox ~]#

To direct input to this VM, move the mouse pointer inside or press Ctrl+G
29°C
Mostly clear
Sep 22 22:33
10:33 PM
9/22/2026
```